

Central Composite Designs

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Introduction

The central composite design (CCD), introduced by Box and Wilson in 1951, is the workhorse of response-surface methodology and arguably the most influential design construction in industrial statistics. It augments a 2^k factorial with axial (or star) points placed along each factor axis at distance α from the design centre and a number of replicated centre points. Together these three components — factorial, axial, centre — provide enough information to fit a full second-order polynomial response surface in k factors with substantially fewer runs than a 3^k full factorial would require, while still supporting estimation of curvature, lack-of-fit testing, and prediction-variance characterisation across the design region.

Prerequisites

A working understanding of two-level factorial designs, polynomial regression for response-surface modelling, and the concept of design rotatability — that prediction variance depends only on radial distance from the centre.

Theory

A CCD has three components:

- **Factorial:** 2^k runs at the corners of the unit cube (± 1 in each factor).
- **Axial:** $2k$ runs at $(\pm\alpha, 0, \dots, 0), (0, \pm\alpha, 0, \dots, 0), \dots, (0, \dots, 0, \pm\alpha)$.
- **Centre:** replicated $(0, 0, \dots, 0)$ runs supplying pure-error estimation.

The choice of α determines the design's properties: rotatable CCDs use $\alpha = \sqrt[4]{2^k}$ (giving constant prediction variance at fixed distance from the centre), face-centred CCDs use $\alpha = 1$ (staying within the original ± 1 cube, useful when factor extremes are constrained), and orthogonal CCDs choose α to make the columns of the second-order design matrix orthogonal.

Assumptions

The response is well modelled by a second-order polynomial within the design region, the residual error is approximately Normal with constant variance, and the centre-point replicates provide a valid pure-error estimate against which lack-of-fit can be tested.

R Implementation

```
library(rsm)

ccd <- ccd(~ x1 + x2, n0 = c(3, 3), alpha = "rotatable",
          randomize = FALSE)
ccd

set.seed(2026)
ccd$y <- 10 + 2 * ccd$x1 + 1.5 * ccd$x2 - 1.2 * ccd$x1^2 -
        0.8 * ccd$x2^2 + 0.6 * ccd$x1 * ccd$x2 +
```

```

rnorm(nrow(ccd), 0, 0.5)

fit <- rsm(y ~ S0(x1, x2), data = ccd)
summary(fit)

steepest(fit, dist = 1.0)

```

Output & Results

`ccd()` constructs the layout, and `rsm()` fits the second-order model with stationary-point analysis, eigenvalue decomposition, and a lack-of-fit test against the pure-error replicates. The `steepest()` function traces the path of steepest ascent, useful when the stationary point lies outside the design region and follow-up experimentation is needed in the optimal direction.

Interpretation

A reporting sentence: “The 14-run rotatable CCD with three centre points identified a concave response surface with stationary point at $(x_1, x_2) = (0.83, 0.69)$, predicted optimum 12.3. Quadratic terms in both factors were significant ($p < 0.01$), confirming curvature; the interaction term was marginal ($p = 0.07$). The lack-of-fit test was non-significant ($p = 0.31$), supporting the second-order model. Path-of-steepest-ascent analysis was unnecessary because the optimum lay inside the design region.” Always run a lack-of-fit test and report the canonical-form classification.

Practical Tips

- Rotatable CCDs give equal prediction variance at fixed distance from the centre and are the default RSM choice; the rotatability property is what makes CCDs ideal for sequential exploration of the response surface.
- Face-centred CCDs ($\alpha = 1$) are useful when factor levels cannot exceed the original cube — common when factor ranges are determined by safety or feasibility limits — at the cost of slightly reduced prediction-variance properties.
- Include 3–5 replicated centre points for pure-error degrees of freedom and curvature testing; fewer than 3 leaves the lack-of-fit test underpowered.
- After fitting, use `contour()` and `persp()` plots to visualise the response surface; engineers and scientists overwhelmingly prefer surface plots to coefficient tables when interpreting RSM results.
- Follow up the second-order fit with canonical analysis to classify the stationary point as a maximum, minimum, or saddle (eigenvalue signs of the quadratic-form matrix); a saddle point usually means the optimum lies outside the design region and additional runs are warranted.
- For more than three factors, consider sequential experimentation: a fractional factorial first to screen for active factors, then a CCD or BBD on the active subset to characterise curvature.

R Packages Used

`rsm::ccd()` and `rsm::rsm()` for canonical CCD construction, second-order fitting, canonical analysis, and steepest-ascent paths; `qualityTools::rsmDesign()` for an alternative interface; `Vdgraph` for variance-dispersion graphs comparing rotatable, face-centred, and orthogonal CCD variants; `daewr` for textbook companion datasets and `DoE.wrapper` for higher-level DoE workflow integration.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans